

A COMPARISON OF THE EFFECT OF SIMPLE AND SUCCUSED DILUTIONS OF ACETYLCHOLINE ON ISOLATED HEART

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ABSTRACT

Homeopathic practitioners have reported the specific effects of various potentized and diluted drugs (Chand, 1995 and Taylor, 1998) since long time. On the basis of this concept they are treating various diseases with confidence and success. However, the scientific basis for this concept is not available in the research literature involving isolated tissues. In the present study, the dilutions of Acetylcholine (ACh) prepared in de-ionized water have been tested on electrical and mechanical activity of isolated rabbit's heart for both simple dilutions (SD) and succussed dilutions (SUD). Results demonstrated that the difference in the effect of SD and SUD was found significant at 9x and 3x for systolic force and at 2x for heart rate, duration of action potential and duration of phase 4. However, both the SD and SUD of ACh could produce opposite to the normal effect in different dilutions on heart rate, systolic force, duration of cardiac action potential and its phase 4. It is concluded that SD and SUD differ in properties and effects on isolated tissue. Further, a drug in both the SD and SUD can change its configuration due to interaction with water and air at its surface (Simson, 1909) to produce opposite effect than its normal one.

INTRODUCTION

The homeopathic system of dilution is very appreciated system for medicine. It not only provides effective treatment for various diseases but also exhibits minimum side effects. This system is based upon the assumption that by increasing the degree of succussed dilution, the potency of drug is also increased (Singh, 1996). Succussion is the process in which solution is put into continuous mechanical shaking for a given time period (Cook, 1997). This process is responsible for a change in the original molecule of drug used and its new molecular structure is supposed to depend upon the kind of solvent used (Barnard, 1965). However, the mechanism and type of dilutions that produce reverse effects in

the body are not yet scientifically acknowledged or experimentally proved on isolated tissues as claimed by homeopathic physicians.

The drug ACh has several actions on heart, which are same as those of vagal stimulation. Thus, it slows the rate and diminishes the force of beat, i.e., the negative inotropic and chronotropic action (Katz, 1992). It has been reported that the ventricles are insensitive to both parasympathetic stimulation and ACh (Loffenholz, 1985) although, cholinergic nerves innervate auricles and conducting tissues of His-purkinjie system in ventricles.

This study has been carried out to investigate the difference between the effect of SD and SUD of ACh on various parameters of

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ventricular mechanical and electrical activities recorded from isolated perfused rabbit's heart.

MATERIALS AND METHODS

Experimental Protocol

Rabbits (*Oryctolagus cuniculus*) of either sex ranging from 1.0-1.5 Kg were used for experiments. These experimental animals were killed by decapitation for immediate isolation of heart. The isolated heart was mounted and perfused through aorta by using Langendorff assembly. The perfusate was oxygenated Krebs's Henseleit Buffer having pH regulated at 37°C. The perfusion was carried out through the aorta at a constant pressure. The composition (mM) of oxygenated Krebs's Henseleit Buffer was 118; NaCl, 4.70; KCl, 1.25; CaCl₂, 1.20; MgSO₄, 25; NaHCO₃ & 11; Glucose.

Preparation of Dilution

In all the experiments, SD and SUD of ACh was used. SD was prepared by dissolving 0.01 gm ACh in 10 ml distilled/de-ionized water. This dilution contains 5.5mM of ACh indicating 10⁻³ dilution or 1x potency from which further dilutions were prepared upto 10⁻¹² dilution or 10x potency, this last dilution of 10x potency contains 5.5 * 10⁻⁹ mM ACh. For the preparation of SUD the diluted drugs from 1x to 10x were shaken for two hours in a mechanical shaker.

Recording of Mechanical and Electrical Activities of Isolated Perfused Heart

The isolated heart was perfused through aorta for coronary perfusion and its ventricular activity was recorded through isotonic transducer on oscillograph (Harvard apparatus Ltd. UK, Cat no. 50-8622). Similarly, the electrical activity i.e., extra cellular action potential was also recorded on another channel of this oscillograph by using fine silver wire electrodes inserted into the ventricular myocardium. The records obtained before and after the infusion of 1 ml of the dilutions (simple and succussed) were used to measure the cardiac parameters, i.e., heart rate, systolic force, duration of action potential and duration

of phase 4 of action potential. These parameters were measured with respect to paper speed of oscillograph and calibration of isotonic transducer and isolated preamplifier of Oscillograph.

Data Analysis

All calculated values of various parameters were then calculated for their average and standard error. The statistical comparison was however made between the effect of simple and succussed dilutions by using students t-test at a significance level of 0.05. In addition, ANOVA was also done to ascertain multiple comparisons among the effects of various SD and SUD ranging from 10⁻³ (1x) to 10⁻¹² (10x) on all parameters.

RESULTS

Results of all the parameters regarding the effect of SD & SUD are presented in Table-1.

Effect on Heart Rate

SD (1x to 10x) of ACh, except 2x, 3x & 4x has been found to decrease the heart rate when compared with their respective controls (Fig. 1). This decrease was found maximum, i.e., 27% of control at 1x. However, this effect was either equal or slightly higher than control at 2x, 3x and 4x of ACh SD, representing reverse, i.e., effect opposite to normal. On the other hand, the SUD of ACh (all dilutions except 4x) has also been found to decrease the heart rate by a maximum of 27% of control. While, at 4x SUD of ACh, the heart rate was almost equal to control. Statistical comparison demonstrated highly significant (P<0.0005) difference between SD and SUD only at 2x.

Effect on Systolic Force

SD (1x to 10x) of ACh, except 8x has been found to increase the systolic force when compared with their respective controls (Fig. 2). This increase was found maximum, i.e., 21% of control at 1x. At 8x, the systolic force was about 5% lesser than control, representing reverse effect, i.e., opposite to normal. The SUD of ACh has also been found to increase the systolic force mostly, by a maximum of

20% of control at 1x. However, 3x, 8x and 9x SUD demonstrated reduction in systolic force being reverse effect of ACh. Statistical comparison demonstrated significant ($P<0.0005$) and ($P<0.05$) difference between SD and SUD at 9x and 3x respectively.

Effect on Duration of Action Potential

SD (1x to 10x) of ACh, except 2x and 5x has been found to increase the duration of action potential as compared to their respective controls (Fig. 3). This increase was found maximum i.e., 7% of control at 10x dilution. At 2x & 5x the duration of action potential was about 2% lesser than control, representing the reverse effect. The SUD of ACh has also been found to increase the duration mostly by a maximum of 15% of control at 1x. However, the SUD of ACh at 4x demonstrated reduction in the duration of action potential being reverse effect of ACh. Statistical comparison between the effect of simple and succussed dilutions on this parameter demonstrated significant ($P<0.005$) difference at 2x only.

Effect on Duration of Phase 4 in Action Potential

SD (1x to 10x) of ACh, except 3x, 4x and 9x has been found to increase the duration of phase 4 (Fig. 4) compared to their respective controls. This increase was found maximum at 1x i.e., 216% of control. At 3x, 4x & 9x the duration of phase 4 was about 2 & 4% lesser than control representing the reverse effect. The SUDs of ACh has also been found to increase the duration of phase 4 mostly by a maximum of 181% of control at 1x. However, a reverse effect on this parameter was observed at 6x & 7x of SUD.

Statistical comparison demonstrated highly significant ($P<0.0005$) difference between SD and SUD at 2x only.

DISCUSSION

The controversial reports on the ability of succussion to change the characteristics of a solution (Jerman, 2001 and Aabel *et al.*, 2001) are tested in the present study, in terms of their

effect on electrical & mechanical performance of isolated heart. Results demonstrated inconsistent differences between SD and SUD of ACh at various potencies ranging from 1x to 10x. This difference in the effect on heart rate, duration of action potential and phase 4 is highly significant ($P<0.0005$) between SD and SUD at 2x.

Such differences represent changes in chemical configuration of the original drug molecule of ACh (Barnard, 1965) and also due to interaction of drug & water molecules on succussion (Barnard, 1965). However, at 3x dilution of ACh the systolic force demonstrated just significant difference when compared between SD and SUD. While, at 9x dilution this difference was highly significant ($P<0.0005$) statistically. It is therefore, suggested that higher the potency the greater its dynamic power (Singh, 1996) obtained through potentization (Taylor, 1998) or succussion (Fischbacher, 1997).

The present study cannot justify above proposal for all dilutions studied. In our opinion, the inconsistent effect of dilution observed from 1x to 10x may be due to non-stability of the change in drug molecule on succussion. Moreover, the controversial reports on differences between SD and SUD may be related with such instability in all types of solutions that were subjected to succussion.

In addition, to the above difference among the effects of SD & SUD of ACh, on isolated heart, another important observation is the effect that was opposite to normal one observed for both the SD and SUD (Fig.1-4). Such opposite or reverse effect of various dilutions has also been reported earlier with the suggestion of a change in the original drug molecule, on succussion (Jerman *et al.*, 2001). In the present study, such effect is observed below 100% control in systolic force, duration of action potential and phase 4 (Fig. 2-4). While, it was above 100% control in heart rate (Fig. 1). The opposite effect has been mentioned as hormesis in the earlier studies reported on dilutions (Jerman *et al.*, 2001).

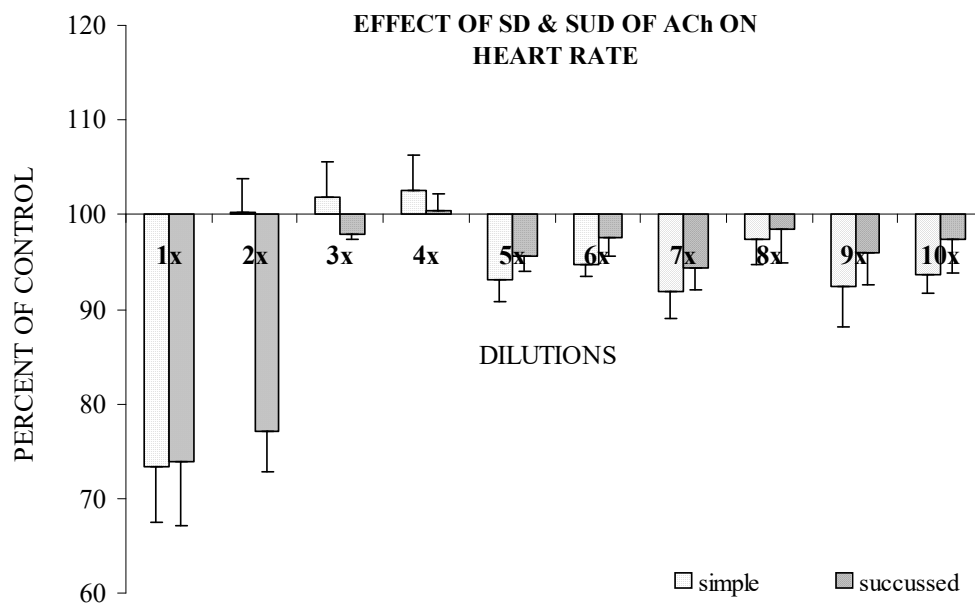


Fig. 1: Effect of SD and SUD of Ach on heart rate.

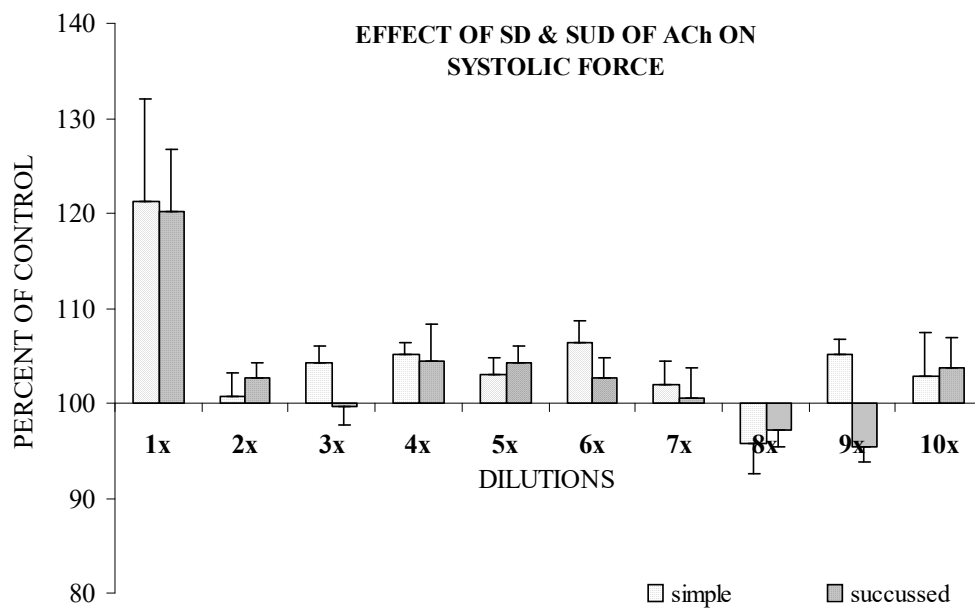


Fig. 2: Effect of SD and SUD of Ach on systolic force.

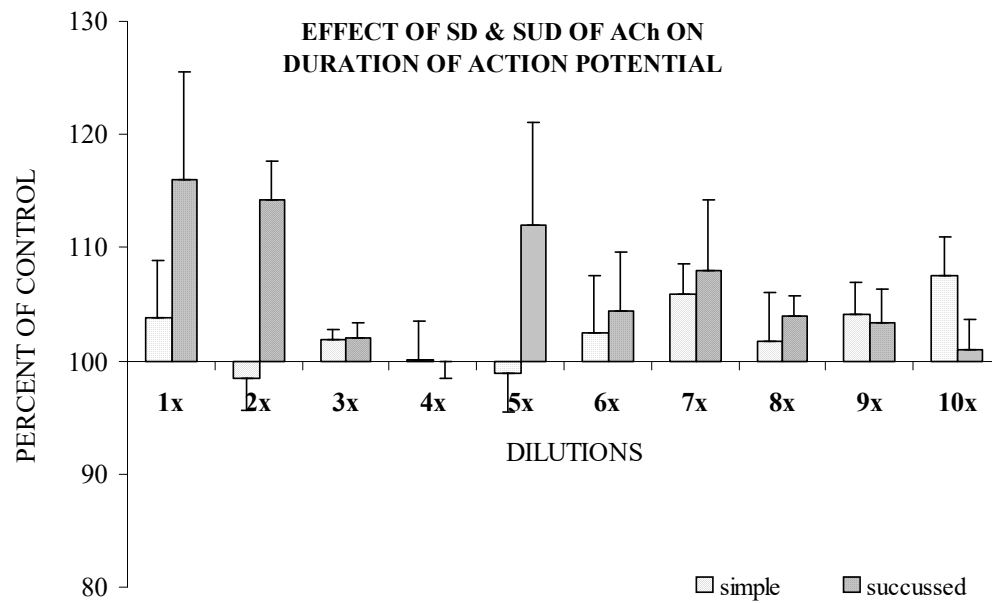


Fig. 3: Effect of SD and SUD of Ach on duration of action potential.

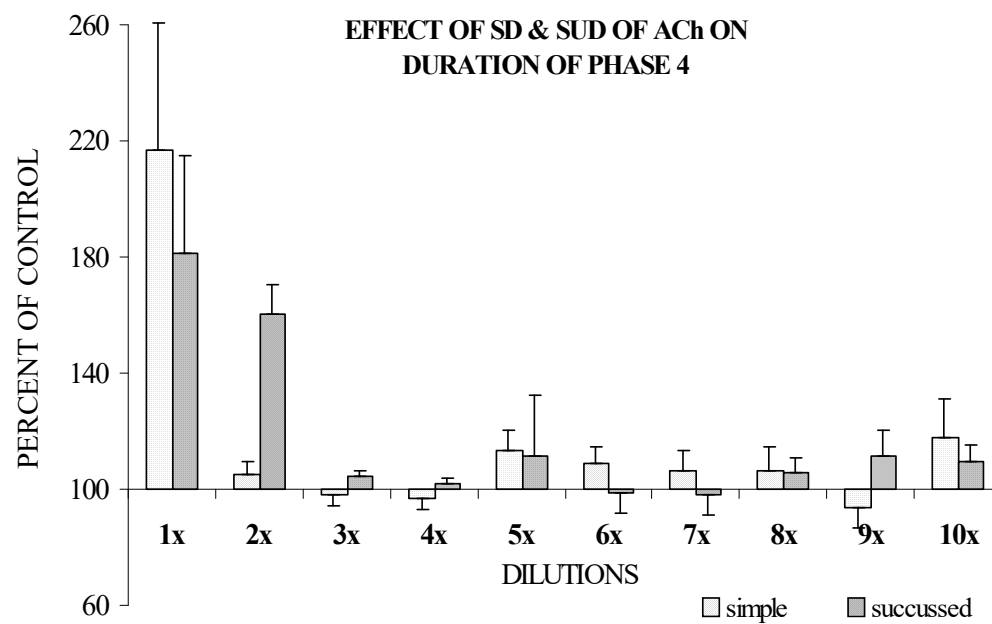


Fig. 4: Effect of SD and SUD of Ach on duration of phase 4.

Table-1
A Comparison between Effect of various SD and SUD of ACh on Electrical and Mechanical
Parameters obtained from Perfused Heart

Parameters	Simple Dilution		Succussed Dilution		P Value
1x / 10-3					
Heart Rate	73.31 ± 5.78	(14)	73.93 ± 6.72	(14)	P>0.05
Systolic Force	121.18 ± 10.9	(14)	120.25 ± 6.41	(14)	P>0.05
Duration of Action Potential	104.25 ± 5.03	(12)	115.44 ± 9.65	(12)	P>0.05
Duration of Phase 4	216.61 ± 43.79	(12)	181.25 ± 33.68	(12)	P>0.05
2x / 10-4					
Heart Rate	*100.29 ± 3.49	(11)	77.17 ± 4.29	(11)	P<0.0005
Systolic Force	100.77 ± 2.47	(11)	102.73 ± 1.54	(11)	P>0.05
Duration of Action Potential	*98.51 ± 2.82	(10)	113.02 ± 3.04	(9)	P<0.005
Duration of Phase 4	104.90 ± 4.61	(10)	159.80 ± 10.18	(9)	P<0.0005
3x / 10-5					
Heart Rate	*101.88 ± 3.62	(11)	97.91 ± 0.61	(24)	P>0.05
Systolic Force	104.30 ± 1.76	(11)	*99.62 ± 1.91	(24)	P<0.05
Duration of Action Potential	101.82 ± 1.00	(10)	102.09 ± 1.24	(20)	P>0.05
Duration of Phase 4	*98.14 ± 3.76	(10)	104.47 ± 1.68	(20)	P>0.05
4x / 10-6					
Heart Rate	*102.53 ± 3.81	(10)	*100.48 ± 1.64	(14)	P>0.05
Systolic Force	105.10 ± 1.34	(10)	104.42 ± 3.94	(14)	P>0.05
Duration of Action Potential	100.03 ± 3.45	(9)	*99.87 ± 1.36	(12)	P>0.05
Duration of Phase 4	*96.78 ± 4.05	(9)	101.72 ± 2.23	(12)	P>0.05
5x / 10-7					
Heart Rate	93.20 ± 2.48	(12)	95.52 ± 1.54	(11)	P>0.05
Systolic Force	103.01 ± 1.79	(12)	104.31 ± 1.68	(11)	P>0.05
Duration of Action Potential	*98.88 ± 3.45	(9)	111.92 ± 9.21	(9)	P>0.05
Duration of Phase 4	113.35 ± 7.10	(9)	111.22 ± 21.34	(9)	P>0.05
6x / 10-8					
Heart Rate	94.79 ± 1.33	(11)	97.65 ± 1.96	(10)	P>0.05
Systolic Force	106.43 ± 2.32	(11)	102.68 ± 2.08	(10)	P>0.05
Duration of Action Potential	102.41 ± 5.11	(13)	104.41 ± 5.16	(8)	P>0.05
Duration of Phase 4	109.00 ± 5.31	(13)	*98.99 ± 7.34	(8)	P>0.05

Table contd...

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Parameters	Simple Dilution		Succussed Dilution		P Value
7x / 10-9					
Heart Rate	91.84 ± 2.9	(12)	94.42 ± 2.45	(11)	P>0.05
Systolic Force	101.98 ± 2.48	(12)	100.59 ± 3.04	(11)	P>0.05
Duration of Action Potential	105.83 ± 2.71	(8)	107.94 ± 6.25	(8)	P>0.05
Duration of Phase 4	106.45 ± 7.02	(8)	*98.09 ± 6.72	(8)	P>0.05
8x / 10-10					
Heart Rate	97.46 ± 2.79	(13)	98.41 ± 3.50	(14)	P>0.05
Systolic Force	*95.78 ± 3.14	(13)	*97.22 ± 1.81	(14)	P>0.05
Duration of Action Potential	101.80 ± 4.29	(10)	102.93 ± 2.35	(9)	P>0.05
Duration of Phase 4	106.33 ± 8.20	(10)	105.97 ± 4.54	(9)	P>0.05
9x / 10-11					
Heart Rate	92.38 ± 4.19	(10)	96.00 ± 3.33	(12)	P>0.05
Systolic Force	105.05 ± 1.74	(10)	*95.32 ± 1.56	(12)	P<0.0005
Duration of Action Potential	104.21 ± 2.57	(10)	103.38 ± 2.98	(10)	P>0.05
Duration of Phase 4	*96.51 ± 6.84	(10)	111.34 ± 8.92	(10)	P>0.05
10x / 10-12					
Heart Rate	93.63 ± 1.95	(12)	97.36 ± 3.60	(13)	P>0.05
Systolic Force	102.75 ± 4.59	(12)	103.74 ± 3.09	(13)	P>0.05
Duration of Action Potential	107.57 ± 3.32	(12)	100.96 ± 2.75	(14)	P>0.05
Duration of Phase 4	118.58 ± 13.01	(12)	109.79 ± 5.15	(14)	P>0.05

All values are presented as mean (% of control) ± standard error (* reverse effect)

CONCLUSION

On the basis of the above discussion it has been concluded that there is a significant difference between some of the SD and SUD in terms of their effect on isolated heart. However, such effect is neither consistent nor stable as not observed at every dilution. Further, both dilutions can produce opposite to the normal effect depending upon the molecular interaction with solvent either due to few jerks in SD or succussion in SUD.

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